

Research Title

Adaptive Learning System Using Artificial Intelligence

Literature Review

1. Introduction

Artificial Intelligence (AI) has emerged as a transformative technology in education by enabling intelligent, personalized, and adaptive learning experiences. Traditional e-learning platforms provide identical learning materials to all students regardless of their individual knowledge level, learning pace, or preferences. In contrast, adaptive learning systems continuously analyse learner performance and dynamically modify learning content, assessments, and recommendations to meet individual learning needs. AI technologies such as machine learning, natural language processing (NLP), recommendation systems, and intelligent tutoring systems play a crucial role in making adaptive learning more effective and engaging.

Recent advances in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have further enhanced adaptive learning by enabling AI tutors to provide context-aware, personalized, and reliable educational assistance based on trusted learning resources.

2. Review of Existing Literature

Gligorea et al. (2023) conducted a comprehensive literature review on artificial intelligence in adaptive e-learning systems by analysing studies published between 2010 and 2023. The authors found that AI significantly improves learner engagement, academic performance, and personalized instruction. Their review highlighted that machine learning algorithms, recommendation systems, learner modelling, and predictive analytics are widely used to personalize educational content. However, the study also pointed out several challenges, including data privacy, interoperability, and the lack of explainable AI models in education.

Wang et al. (2024) performed a meta-analysis involving 45 independent studies to evaluate the effectiveness of AI-enabled adaptive learning systems. Their findings demonstrated that adaptive learning technologies positively influence student achievement, motivation, and knowledge retention. The researchers concluded that adaptive systems capable of continuously monitoring learner performance and dynamically updating instructional content provide better educational outcomes than conventional e-learning systems.

Meylani (2024) presented a systematic review focusing on recent developments in AI-powered adaptive learning systems. The study discussed the increasing adoption of intelligent tutoring systems, personalized learning environments, adaptive assessments, and learning analytics. The review emphasized that future adaptive learning platforms should focus on transparency, explainable AI, learner privacy, and trusted educational resources to improve user confidence and system effectiveness.

A systematic mapping study published in *Computers and Education: Artificial Intelligence* analysed 147 research papers related to adaptive learning technologies. The researchers classified adaptive

learning approaches into learner modelling, intelligent tutoring, recommendation systems, automated assessment, and predictive learning analytics. The study observed that while many adaptive algorithms have been proposed, only a limited number have been implemented as scalable educational platforms suitable for real-world deployment.

Recent studies have also explored Retrieval-Augmented Generation (RAG) as a promising approach for educational AI systems. Instead of relying solely on the internal knowledge of large language models, RAG retrieves relevant information from trusted educational resources before generating responses. This approach improves factual accuracy, reduces AI hallucinations, and provides students with reliable explanations directly linked to course materials. As a result, RAG has become an important component of modern AI-powered tutoring systems.

3. Research Gap

Although numerous studies have explored adaptive learning technologies, several limitations remain in existing systems:

- Most platforms focus only on personalized content recommendation and lack comprehensive adaptive learning capabilities.
- Many adaptive learning systems do not integrate intelligent tutoring, adaptive assessments, learning analytics, and personalized learning paths into a single platform.
- Existing AI tutors often rely solely on large language models, which may produce inaccurate or hallucinated responses.
- Few systems utilize Retrieval-Augmented Generation (RAG) to provide trustworthy responses based on verified educational resources.
- There is limited research on combining learner assessment, continuous performance monitoring, adaptive recommendations, and AI tutoring within one integrated learning platform.

4. Proposed Research

The proposed research aims to design and develop an **AI-powered Adaptive Learning System** that delivers personalized educational experiences based on each learner's knowledge level and learning progress. The system will incorporate artificial intelligence to generate individualized learning paths, provide AI-assisted tutoring, conduct adaptive assessments, analyse student performance, and continuously update learning recommendations.

Unlike traditional e-learning systems, the proposed platform will integrate Retrieval-Augmented Generation (RAG) to ensure that AI-generated responses are derived from trusted educational resources uploaded by course administrators. This approach enhances response accuracy while minimizing misinformation. The system will also include learning analytics to monitor student progress and dynamically modify learning paths according to individual performance.

The proposed solution aims to improve learning effectiveness, student engagement, and knowledge retention by providing a highly personalized and intelligent educational environment.

References (APA 7th Edition)

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